

SHIPRA

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I am currently pursuing **M.Sc. in Data Science and BIOAI** at Digital University Kerala with a practical knowledge of Python programming, machine learning, and deep learning.

Skilled in data analysis, preprocessing, and visualization.

Seeking Data Science and ML Engineer roles where I can apply AI/ML techniques to build predictive models and efficient data pipelines.

EDUCATION

M.Sc. Data Science and BIOAI | Digital University Kerala 2025 – Present

- Coursework: Next Generation Sequencing, Bioinformatics, Molecular Biology, Python Programming, Predictive Analytics, Deep Learning, Generative AI, Big data technologies and Cloud Computing, Advanced programming, Data Engineering, Healthcare Analytics, AI for Molecular Sciences

Bachelor of Science in Botany | Jananayak Chandrashekhar University / Satish Chandra College, Ballia 2022 – 2025

- Completed degree with a foundation in biological sciences, complemented by self-directed study in Python and data analysis

TECHNICAL SKILLS

- Languages & Libraries: **Python, BioPython, scikit-learn, TensorFlow/Keras, Flask, HTML/CSS/JavaScript**
- Data Science: **numpy, pandas, matplotlib, tensorflow, pytorch, keras, linear regression**
- Data Visualization: **Tableau, Power BI, Looker Studio**
- Machine Learning / Deep Learning: **CNN, Neural Networks, Predictive Modeling, Deep Learning Pipelines**
- Bioinformatics Tools: **Galaxy (Usegalaxy), Tablet, NCBI, scikit-bio, Cytoscape, String**
- Cloud & Databases: **Google Cloud Platform, BigQuery, SQL, Data Warehouse, Cloud Storage**
- Other Tools: **Git, GitHub, Jupyter Notebook, VS Code**

PROJECTS

A deep learning and large language hybrid workflow for Omics interpretation | June 2026 - present

Developing a hybrid deep learning–LLM pipeline to identify autophagy regulators in Idiopathic Pulmonary Fibrosis (IPF), adapting the LyMOI framework (<https://doi.org/10.1038/s41551-025-01576-5>) using multi-omics data, protein-protein interaction graphs, and LLM-based reasoning.

Attention-Guided Neural ODE Network for Breast Tumor Segmentation | Feb 2026- May 2026

- Applied deep learning to identify and segment tumors in ultrasound images using attention-guided neural ODE architecture
- Implemented CNN-based model in Python/TensorFlow; contributed to advancing medical image analysis using AI

RNA-Seq Differential Gene Expression Analysis for Lung Cancer Using Galaxy | March 2026 - May 2026

Tools: Galaxy, FastQC, Cutadapt/Trim Galore, HISAT2 (or STAR), featureCounts/HTSeq, DESeq2, GEO, NCBI SRA

Conducted RNA-seq analysis of normal and tumor lung tissue samples from the GEO dataset GSE81089 using the Galaxy platform. Implemented quality assessment (FastQC), preprocessing, read alignment, gene counting, differential expression analysis, and functional enrichment to identify potential biomarkers and dysregulated pathways.

CERTIFICATIONS & COURSES

- (Udemy, Certificate ID: [UC-11cb101c-0174-4c9e-b2a6-07f568a6c316](#)) AI & Python Development – 300+ Hands-On Projects
- 2 days hands-on workshop on Molecular Docking by Bioture
- 2 days hands-on workshop on Network Analysis using Cytoscape by Bioture
- Google Cloud Data Analytics Certificate- https://www.skills.google/public_profiles/fb1cf2b0-4572-4c7c-9467-33a005f0cb06
- Introduction to Git and GitHub by Google- https://www.coursera.org/account/accomplishments/verify/OC3UQHB1ZN1M?utm_source=link&utm_medium=certificate&utm_content=cert_image&utm_campaign=sharing_cta&utm_product=course